air–sea CO_2 flux [mmol C m^{-3} s^{-1}] (>0 outgassing) — distance × time (2006–2023, daily)



Upstream (river) boundaries are CONSTANT for every solute except T and TOC -- same limitation as the original 44-yr record. Marine (mouth) boundaries, by contrast, are genuinely time-varying here for EVERY solute (S, DIC, ALK, NO3, NH4, PO4, dSi, O2, TOC -- ECCO-Darwin v5, 2005-2023), which is new relative to the original figures (only T was marine-time-varying there). Structure seen here reflects BOTH the model own internal dynamics acting on a constant upstream source AND a genuinely time-varying ocean signal propagating in from the mouth. See CLAUDE.md -> "Interannual forcing" -> "Full-forcing interannual (2005-2023)".